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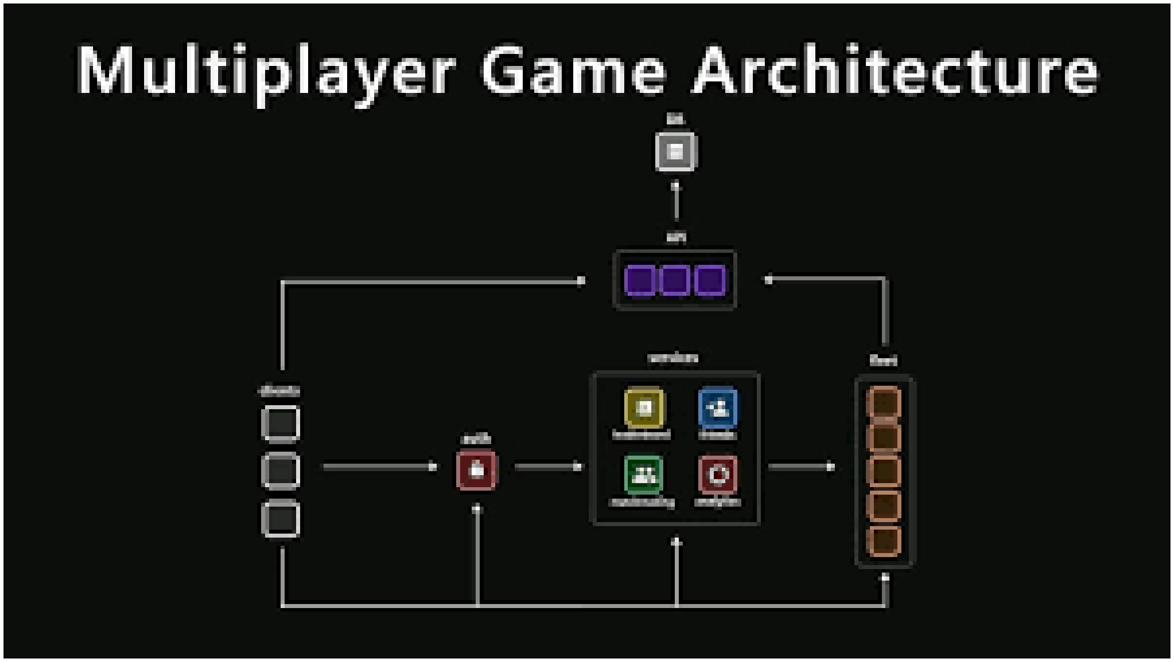
CA-NETWORK (CA-N)

ASSIGNMENT – 2

Identify a real-world application for both parallel computing and networked systems. Explain how these technologies are used and why they are important in that context.


Multiplayer Game Architecture
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A Real-World Application : Online Multiplayer Gaming.

Parallel Computing in Gaming

In online multiplayer games, parallel computing is used for:

Rendering Graphics: High-quality 3D graphics are rendered in real time using multiple processors or cores (e.g., GPUs).

Plastic and AI Calculations: simulating realistic physics (collisions, motion) and complex AI for non-players characters (NPCs)

Concurrent Processes: Running game logic, input handling, and audio processing simultaneously improves responsiveness and realism.

Importance: Enhance user experience with smooth, real-time ~~interaction~~ interactivity and detailed environments.

Network systems in Gaming

Networked Systems are vital for:

Connecting Players: Players connect over the internet or LAN to participate in shared game sessions

Client - Server Model: A central server manages game state and synchronizes actions across all players.

Real-Time Communication: Data like position updates, chat, and events are constantly exchanged with minimal delay.

Importance : Ensures fair play, data consistency, and enables global interaction among players.

Why These Technologies Matter

Scalability : Both technologies enables the game to support large number of users simultaneously

Efficiency : Workload distribution (parallel computing) and seamless connectivity (networking) optimize performance.

User Engagement : Realistic, lag-free gameplay enhances player satisfaction and retention.